1. A 5-methyl guanosine base is added to the 5' end of the transcript
2. The 3' end is added with poly Adenine bases known as the poly A tail.
3. The eukaryotic genes and there by their corresponding RNA transcripts can have multiple non coding regions in between known as *introns*. The coding regions are known as exons. The process by which the noncoding *intron* regions are removed from the RNA transcript is known as RNA splicing. This processed RNA molecule which is ready for translation is known as an mRNA or a messenger RNA.

6.3.5 Difference between Prokaryotic and Eukaryotic in Transcription

Sl. No	Prokaryotic Transcription	Eukaryotic Transcription
1	Coupled transcription-translation is the rule	Coupled transcription translation is not possible.
2	Occurs in the cytoplasm	Occurs in the nucleus
3	There is no definite phase for occurrence	Majority of transcription takes place in the G1 and G2 phase of the cell cycle
4	A single RNA polymerase synthesis all the three types of RNA(mRNA,tRNA,rRNA)	Different types RNA polymerases (I,II,III etc) synthesize different RNA molecules such as rRNA,mRNA,tRNA etc...
5	Initiation of transcription factor does not need any proteins or initiation factors	Initiation of transcription proteins requires transcription factors.These are TFIIA,TFIIB,TFIID,TFIIE,TFIIH.These recognises the TATA box.
6	Transcription usually involves more than one gene (polycistronic)	Transcription normally involves only one gene (monocistronic)
7	Usually there is no post transcriptional modification of the primary transcript.	Primary transcript undergo post translational modification(RNA editing)
8	Introns absent in the primary transcript.	Introns present in the primary transcript.

6.4 TRANSLATION

As we have already studied, Proteins are the key molecules which literally carry out every single molecular process inside our cell. The instructions about how these proteins should be made is encoded in the DNA and is transcribed from the chromosome in the form RNA by a process which we saw as transcription. Similalry the process by which a new protein molecule is synthesized using this mRNA as a template is known as translation.

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